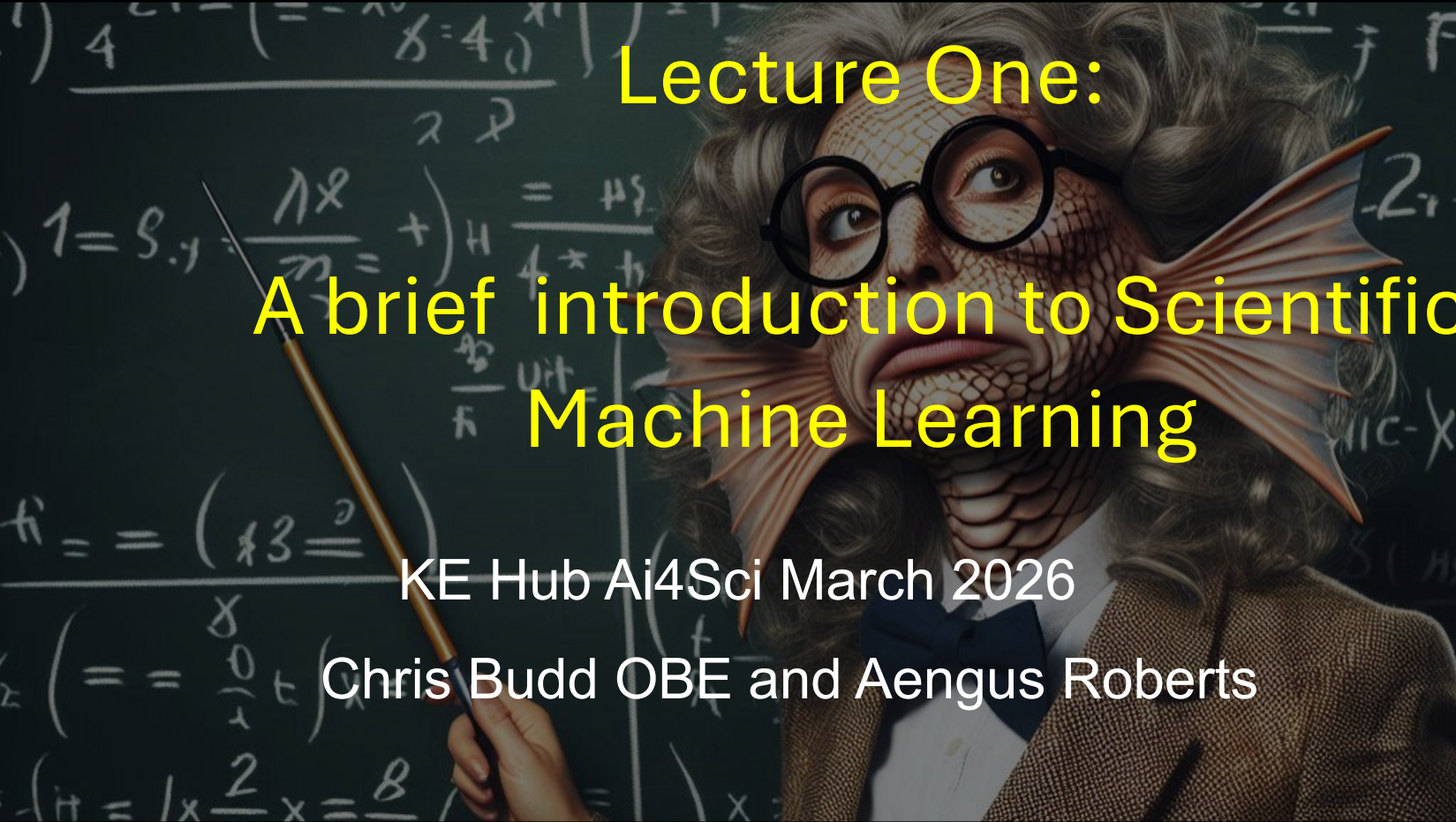
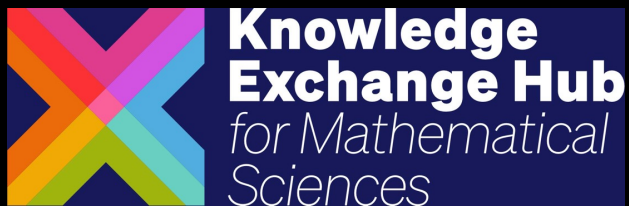




Lecture One: A brief introduction to Scientific Machine Learning

KE Hub Ai4Sci March 2026

Chris Budd OBE and Aengus Roberts



www.bbc.co.uk/news/science-environment-67383755

NEWS

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Science & Environment

AI could predict hurricane landfall sooner - report

🕒 14 November 2023 · 💬 45 Comments

🔗

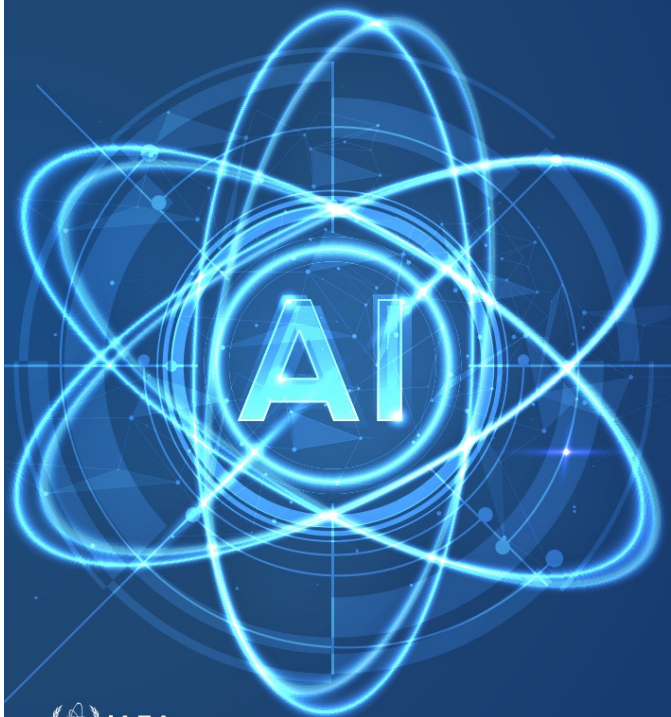


AI In the News

Makes 'better prediction of Hurricane Lee' than any other method

How good is the hype?

**Artificial Intelligence for
Accelerating Nuclear Applications,
Science and Technology**



Deep Learning



If you were ill, would you trust a diagnosis only made by a machine without involving a doctor?

Would you trust a weather forecast made by a machine without using the laws of physics?



Would you be happy to fly on aeroplane designed just from looking at previous aeroplanes?

All of the above are possible now, or in the near future, thanks to machine learning. But should we trust them?

Poster by
Helena
Lake

Structure of the mini-course

1. Introduction

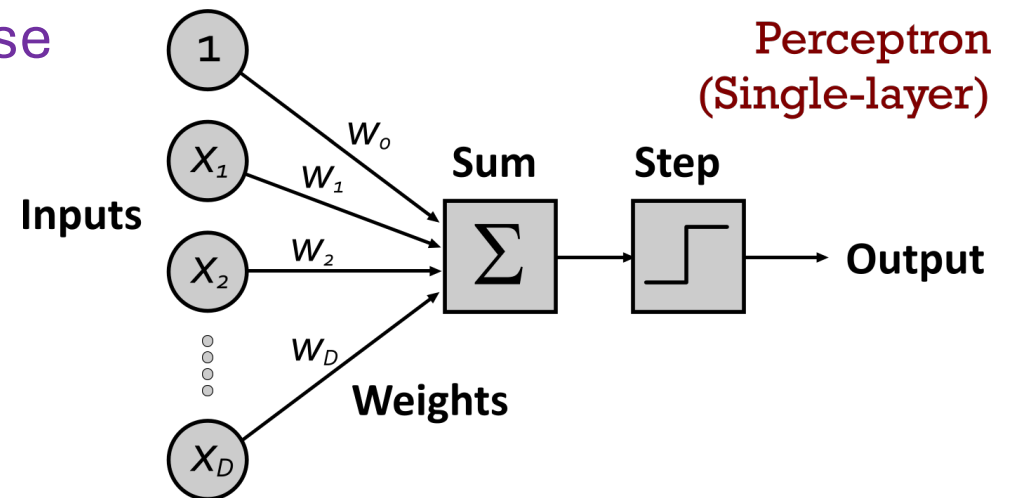
2. Architectures

3. ML for differential equations

4. Neural operators and the weather

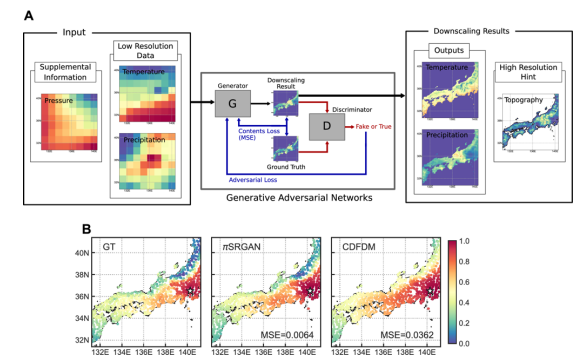
5. Neural ODES

6 Case Study: PINNS



We are seeing a major impact of machine learning mathematics, physics, engineering, and business

- Classification and identification
- Solution of differential equations (ODEs and PDEs)
- Simulation of differential operators
- Solution and regularization of inverse problems
- Prediction and time series analysis
- Classification and generation of images (down scaling)
- CAD
- Forecasting the weather (from data alone!)



Physics



Conventional
modelling



Compare
predictions
with data

Training Data



AI Black Box



Compare
predictions
with physics

Data and
physics



Scientific (physics
informed) machine
learning



Compare
predictions
with physics
and data

Scientific Machine Learning

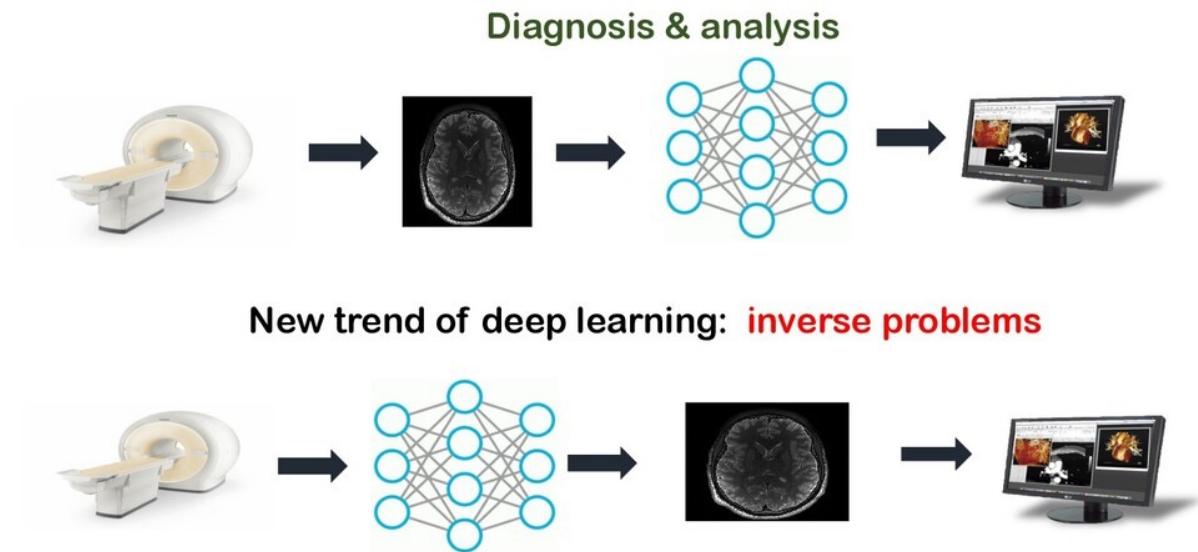
Aims:

- Derive
 - Effective
 - Efficient
 - Transparent
 - Reliable
 - Unbiased (data, inductive, learning)

Machine learning methods informed by physics and other disciplines

- Prove (where possible) rigorous results on their accuracy, stability, and convergence
- Apply them to reliably solve **important scientific and engineering problems**

Deep Learning for Inverse Problems



- A brief introduction
to the world of
(scientific) machine
learning



Image classification/Prediction

Probability distribution of images

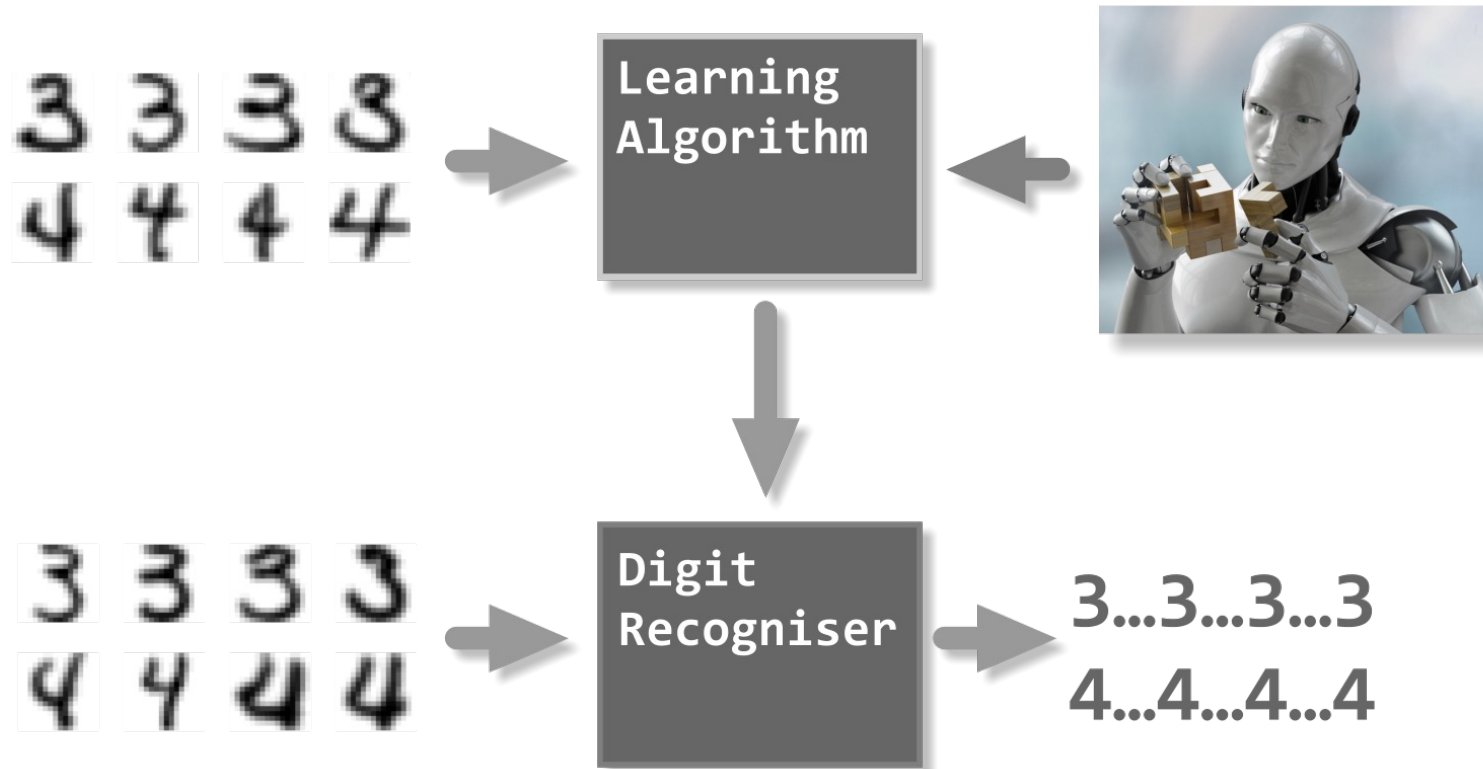


Neural net trained on
samples from the set of
images



Image labels

Operation of a classifier





shutterstock.com • 1046217481

Example: Can you tell a cat from a dog?

Feature identification

One measurement is not good enough

Make two
measurements

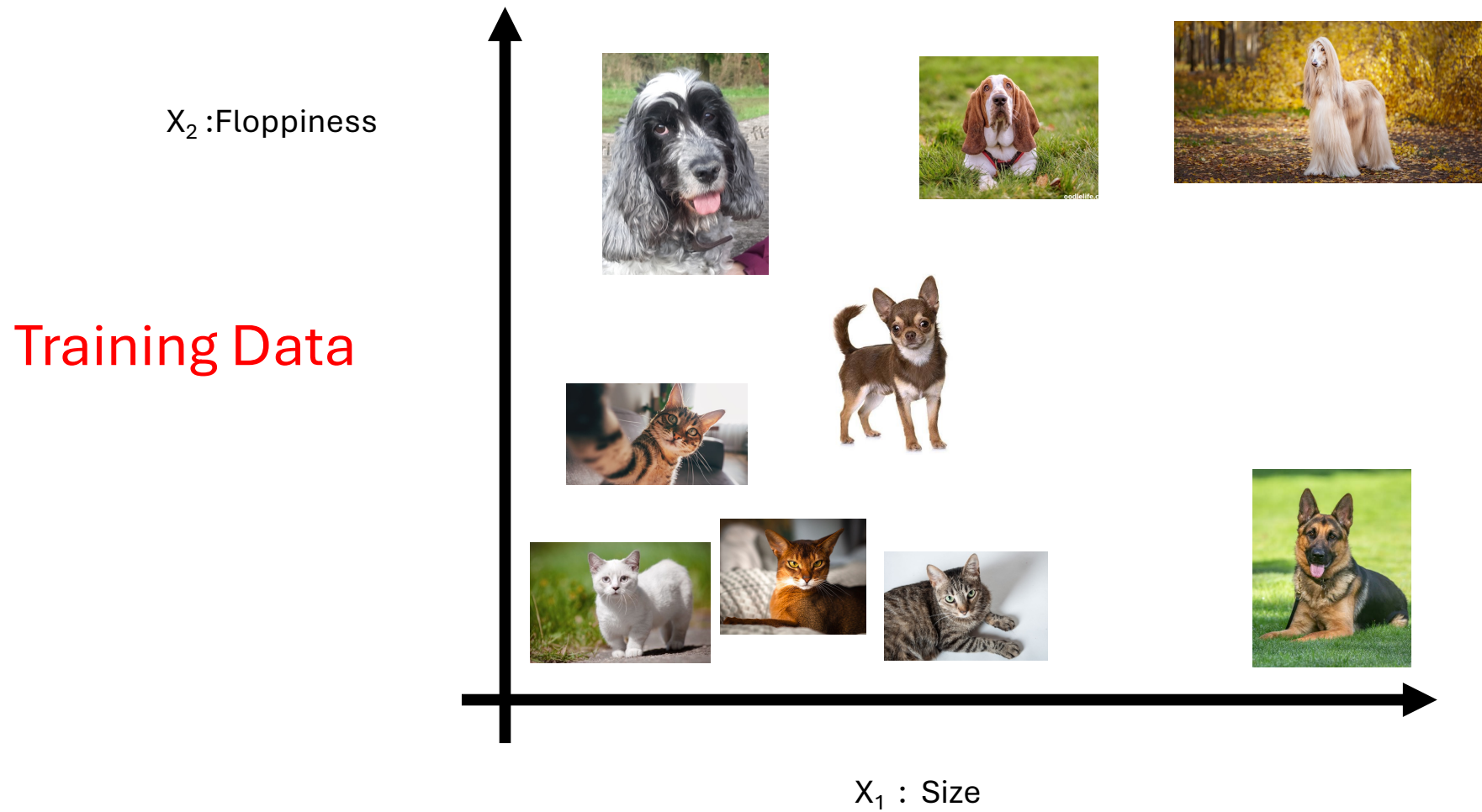


X_1 : Size

X_2 : Floppiness



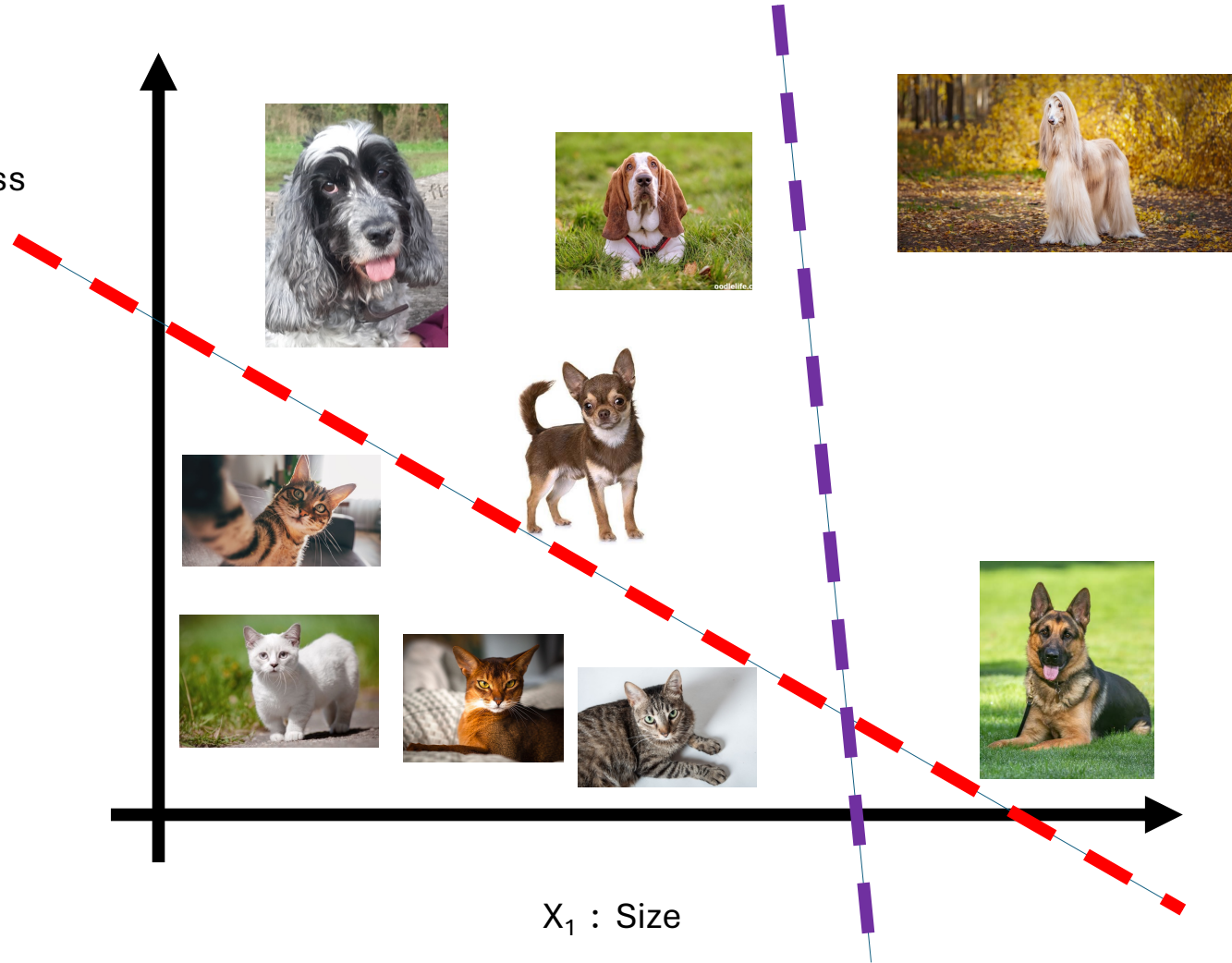
Measure size and floppiness for lots of cats and dogs and plot



Draw a line which separates cats from dogs

Training the
classifier

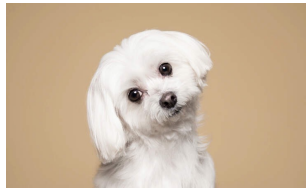
X_2 : Floppiness



Can now tell a cat from a dog

Using the
classifier

X_2 : Floppiness



Dog



Cat

X_1 : Size

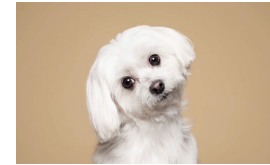
What is going on?

Equation for the line is:

$$w_1X_1 + w_2X_2 - C = 0$$

$$w_1X_1 + w_2X_2 - C > 0 : \text{Dog}$$

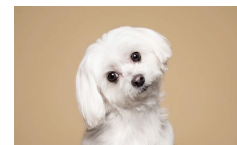
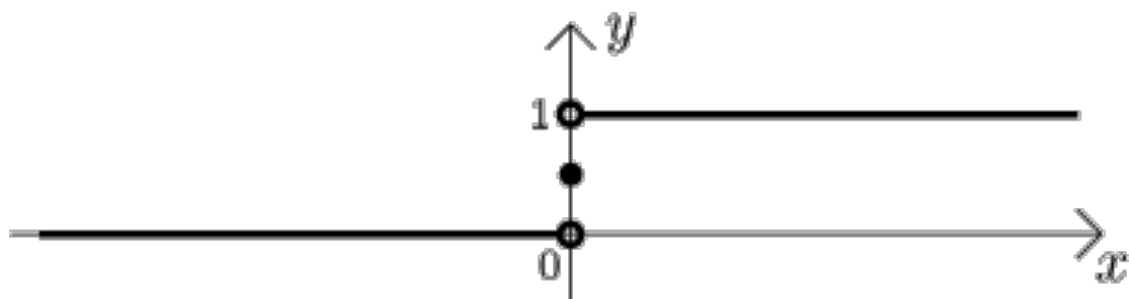
$$w_1X_1 + w_2X_2 - C < 0 : \text{Cat}$$



Train the **weights** w_1, w_2, C on the data to give the right result

$$Out = H \left(\sum_{i=1}^D w_i X_i - C \right)$$

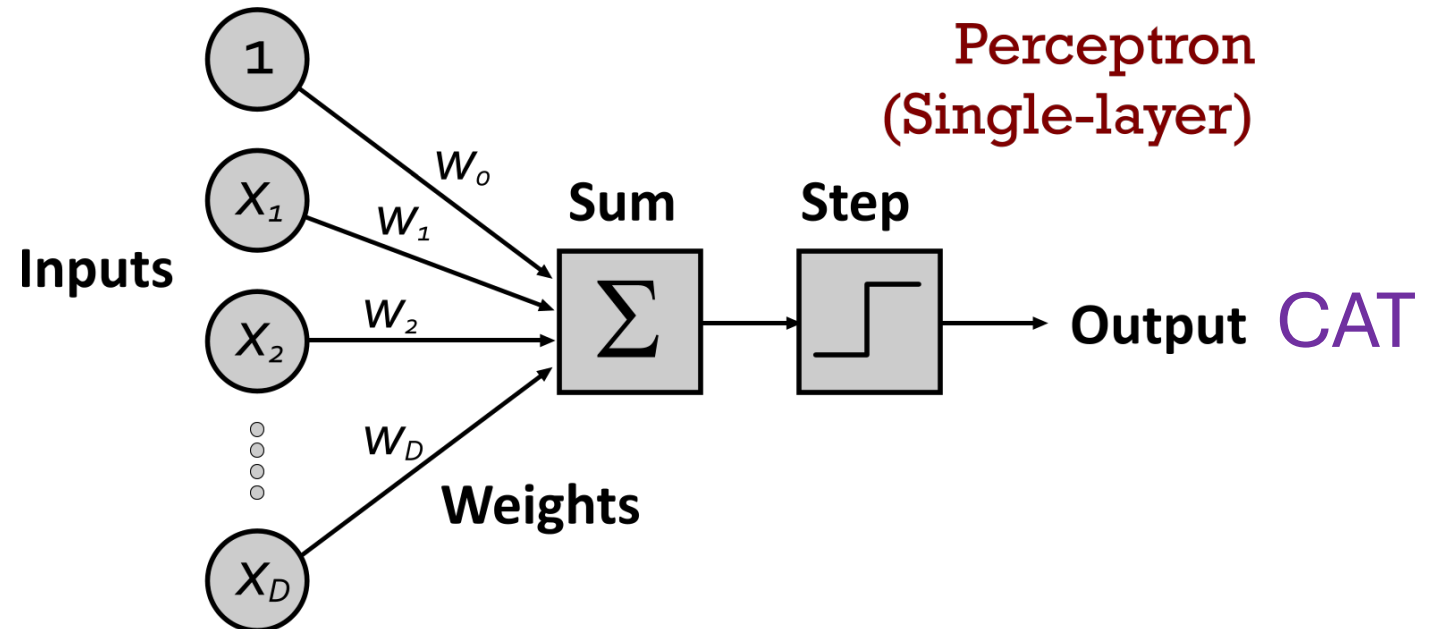
$$H(x) = \begin{cases} 1, & x > 0; \\ \frac{1}{2}, & x = 0; \\ 0, & x < 0. \end{cases}$$



Extract important features

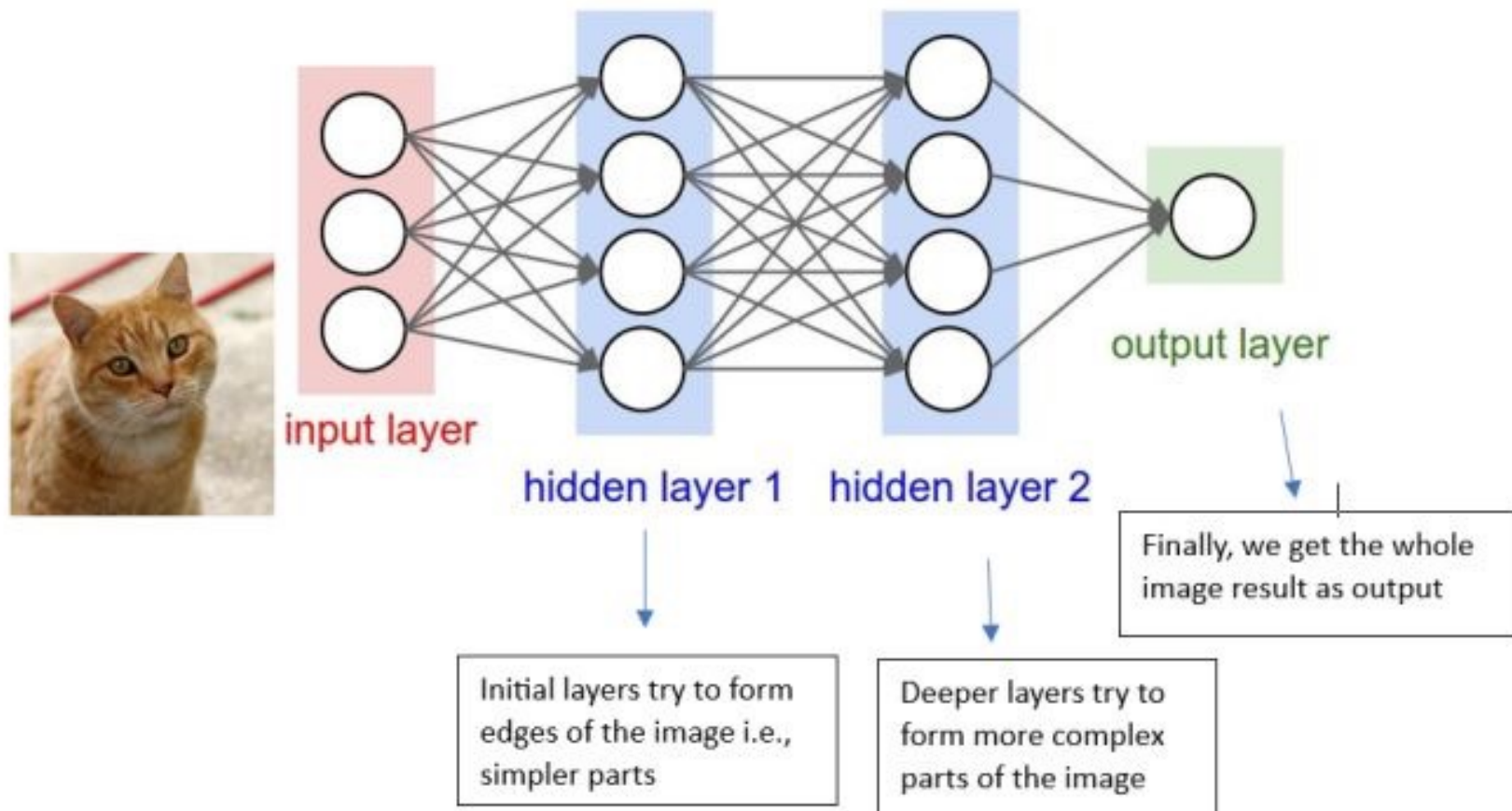


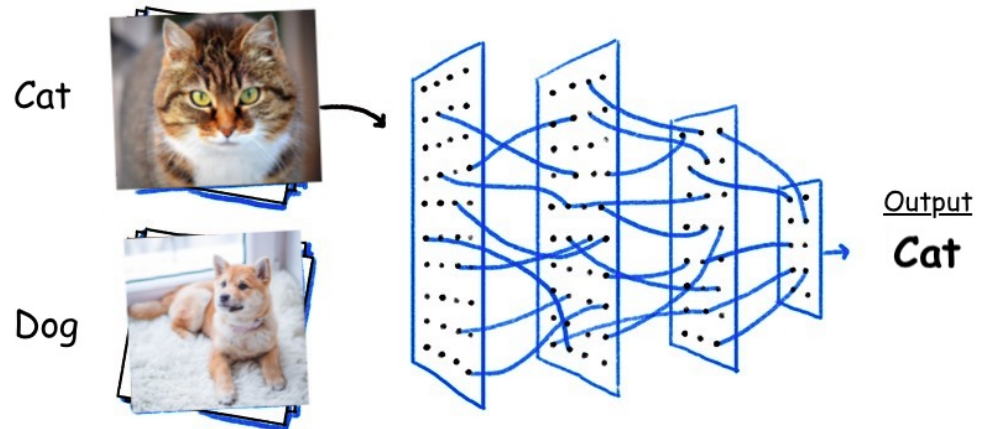
Image



Basic architecture of a neural net classifier (1963)

Deep neural nets (2010) use many more layers and can be used to make much more complex decisions





Or more mathematically, using a notation we use later on:

$$y_{j+1} = \sum_{i=0}^{W-1} c_i \sigma_{i,j} (a_{i,j} y_j + b_{i,j})$$

Input: $y_0 = x$. Output: $y(x) = y_L$

W: Width L: Depth

$\sigma_{i,j}$: Activation functions. (usually nonlinear)

θ : Set of All of the parameters

Name	Plot	Equation	Derivative
Identity		$f(x) = x$	$f'(x) = 1$
Binary step		$f(x) = \begin{cases} 0 & \text{for } x < 0 \\ 1 & \text{for } x \geq 0 \end{cases}$	$f'(x) = \begin{cases} 0 & \text{for } x \neq 0 \\ ? & \text{for } x = 0 \end{cases}$
Logistic (a.k.a Soft step)		$f(x) = \frac{1}{1 + e^{-x}}$	$f'(x) = f(x)(1 - f(x))$
TanH		$f(x) = \tanh(x) = \frac{2}{1 + e^{-2x}} - 1$	$f'(x) = 1 - f(x)^2$
ArcTan		$f(x) = \tan^{-1}(x)$	$f'(x) = \frac{1}{x^2 + 1}$
Rectified Linear Unit (ReLU)		$f(x) = \begin{cases} 0 & \text{for } x < 0 \\ x & \text{for } x \geq 0 \end{cases}$	$f'(x) = \begin{cases} 0 & \text{for } x < 0 \\ 1 & \text{for } x \geq 0 \end{cases}$
Parametric Rectified Linear Unit (PReLU) [2]		$f(x) = \begin{cases} \alpha x & \text{for } x < 0 \\ x & \text{for } x \geq 0 \end{cases}$	$f'(x) = \begin{cases} \alpha & \text{for } x < 0 \\ 1 & \text{for } x \geq 0 \end{cases}$
Exponential Linear Unit (ELU) [3]		$f(x) = \begin{cases} \alpha(e^x - 1) & \text{for } x < 0 \\ x & \text{for } x \geq 0 \end{cases}$	$f'(x) = \begin{cases} f(x) + \alpha & \text{for } x < 0 \\ 1 & \text{for } x \geq 0 \end{cases}$
SoftPlus		$f(x) = \log_e(1 + e^x)$	$f'(x) = \frac{1}{1 + e^{-x}}$

Need to find the best values for the Parameters

Achieve this by training the NN using a training set of data and a measure of success

Can take many hours of computer time on a GPU.

With a vast expense of energy!!



- Supervised training

Have a **labeled training set**

eg. cats + dogs, musical genres,
Inverse problems, approximation

- Reinforcement training

Have a measure of success

eg. winning GO, solving an equation
(eg. a PINN), variational method



Mathematical Ingredients of the training process

Loss function $L(W)$: How close is the answer

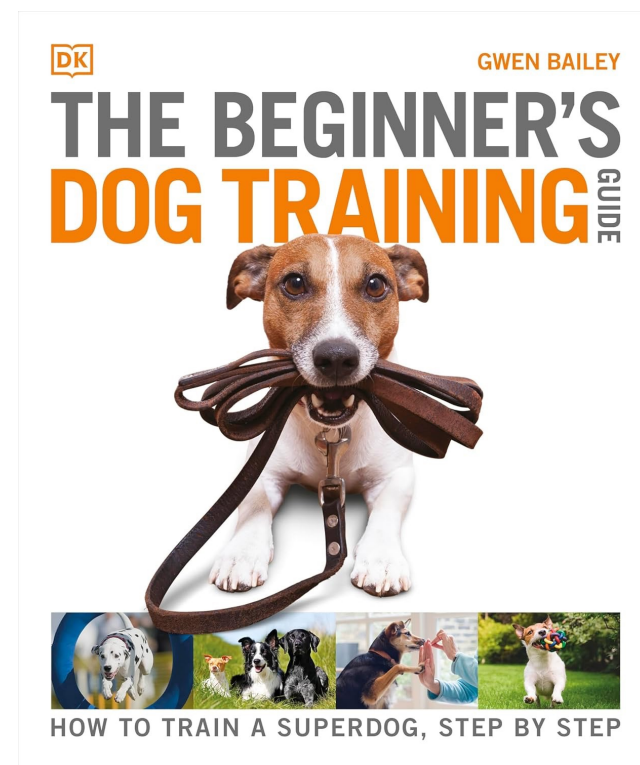
Back propagation: How does the answer depend on changes in the parameters.

Achieved by using automatic differentiation

PyTorch does this with ease!

Optimisation: Choose the parameters to minimize the loss over (a subset of the) training data

Validation: Test on separate data to avoid over fitting to the answer



Optimise the weights using optimisation methods:

Eg. stochastic gradient descent, ADAM, quasi-Newton, simulated annealing, Monte-Carlo Tree Search (MCTS)

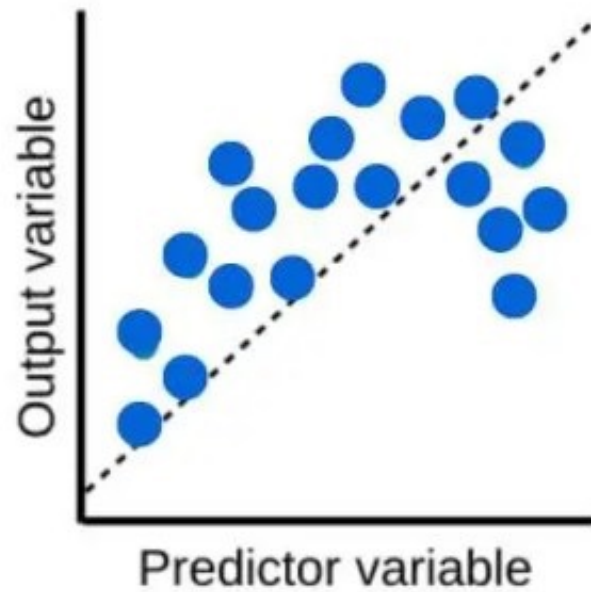
Implemented in eg. PyTorch
(see the Case studies)



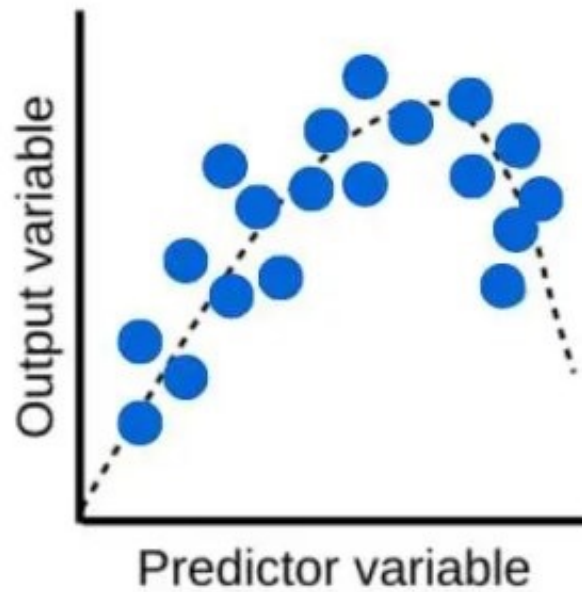
Tensor Flow/Keras

AlphaZero used reinforcement learning over 700,000 games with MCTS optimisation

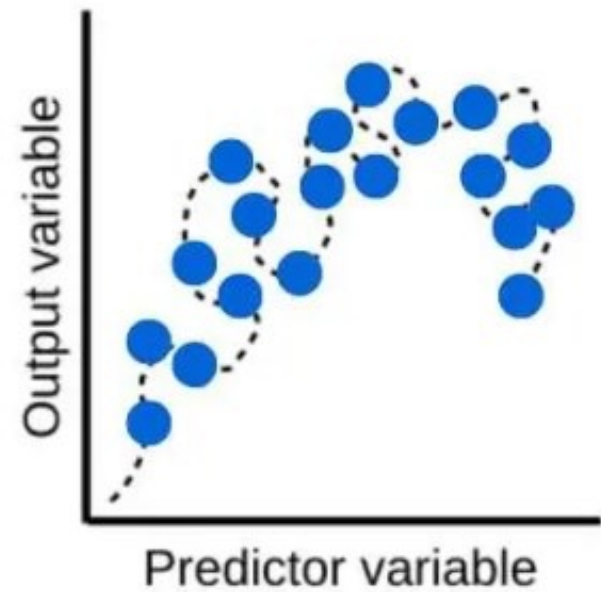
Underfitting

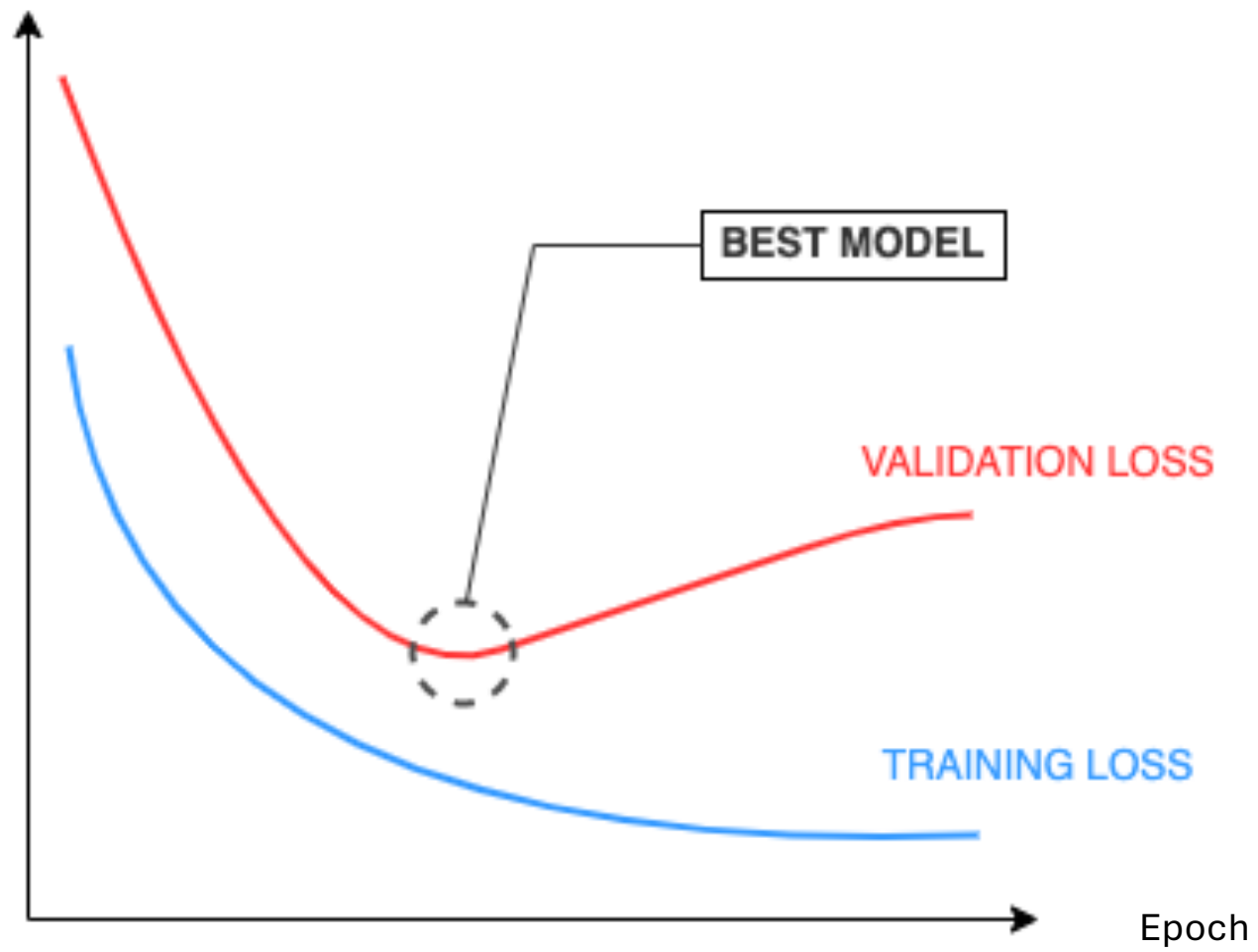


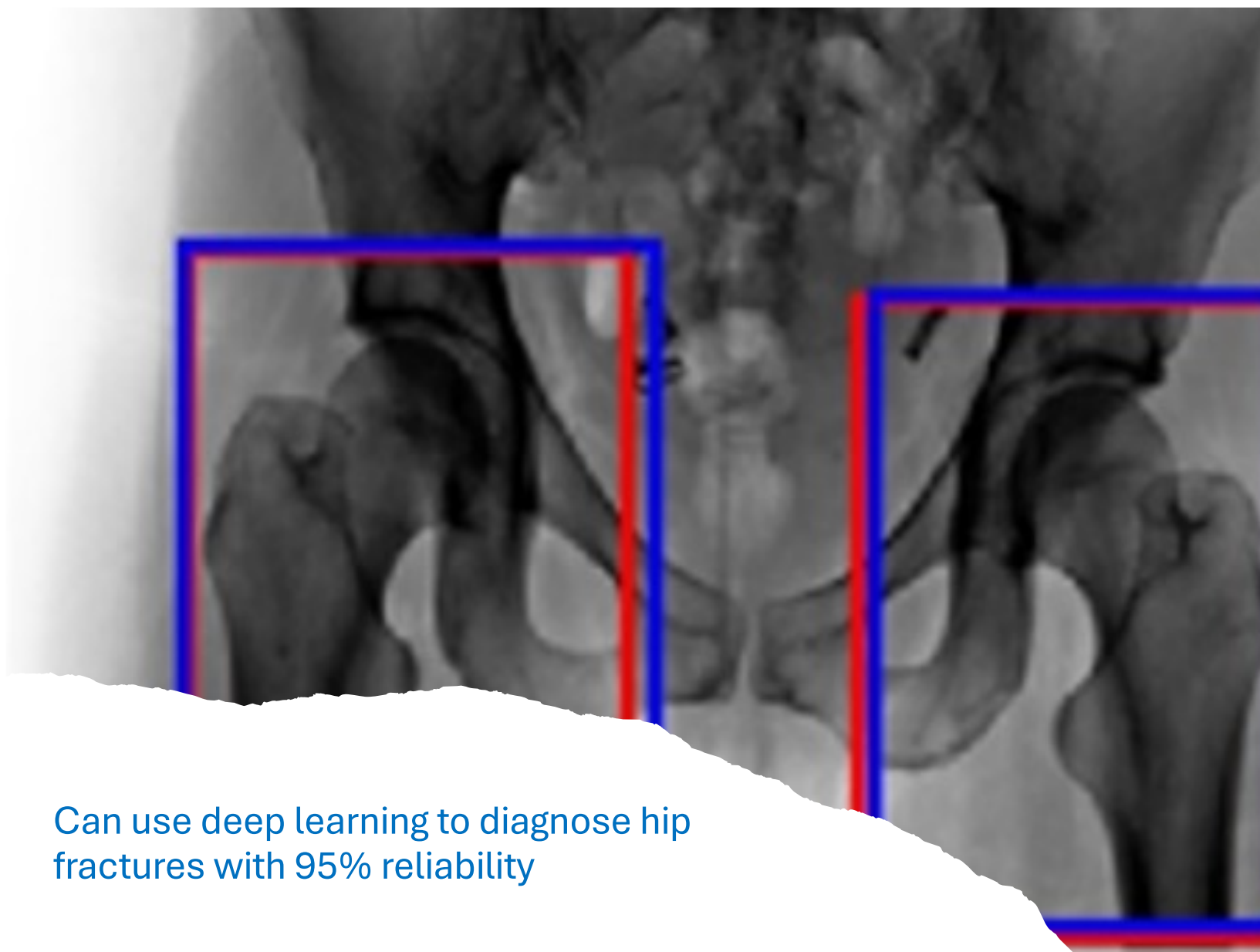
Optimised Fit



Overfitting

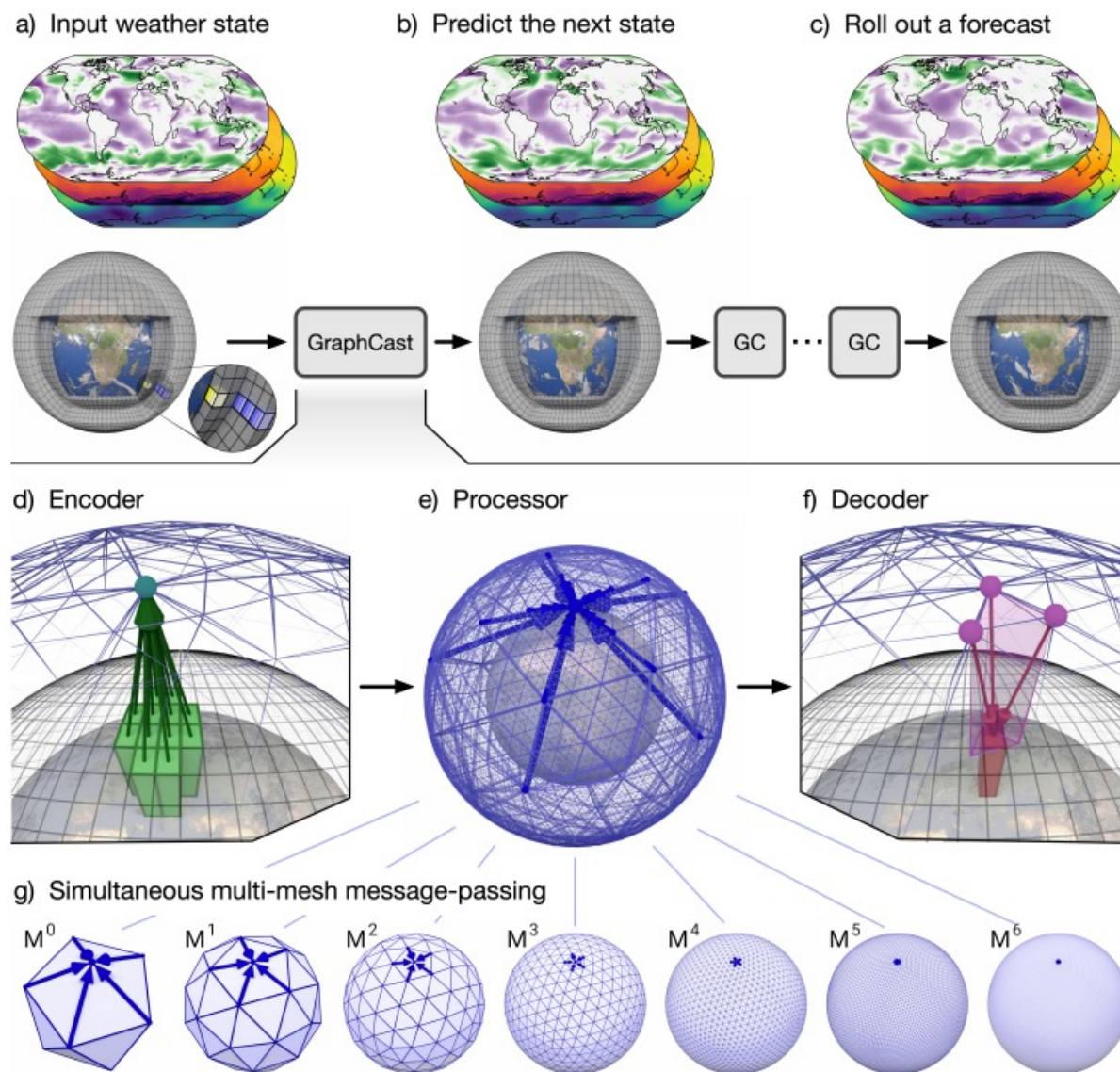






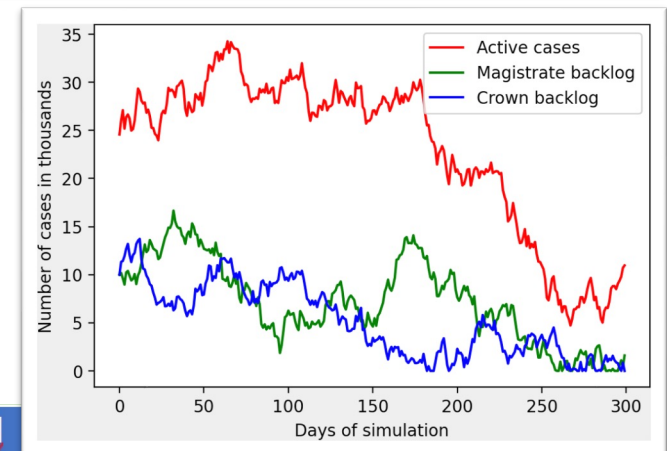
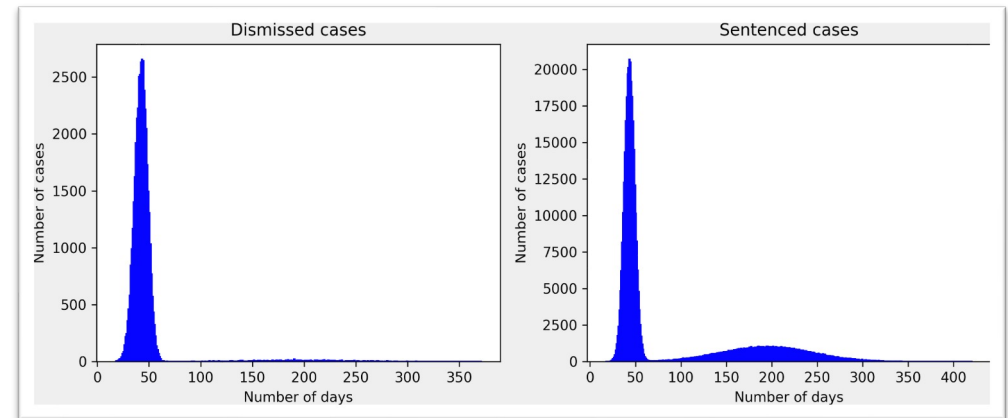
Can use deep learning to diagnose hip fractures with 95% reliability

Weather forecasting trained on the ERA5 Data Set



Maths and AI for Justice

NN can predict case volume and duration in the judicial system based on case characteristics



Time distribution: Lognormal

Time spent in summoned: 34 days

Summoned to Exit: 9 days

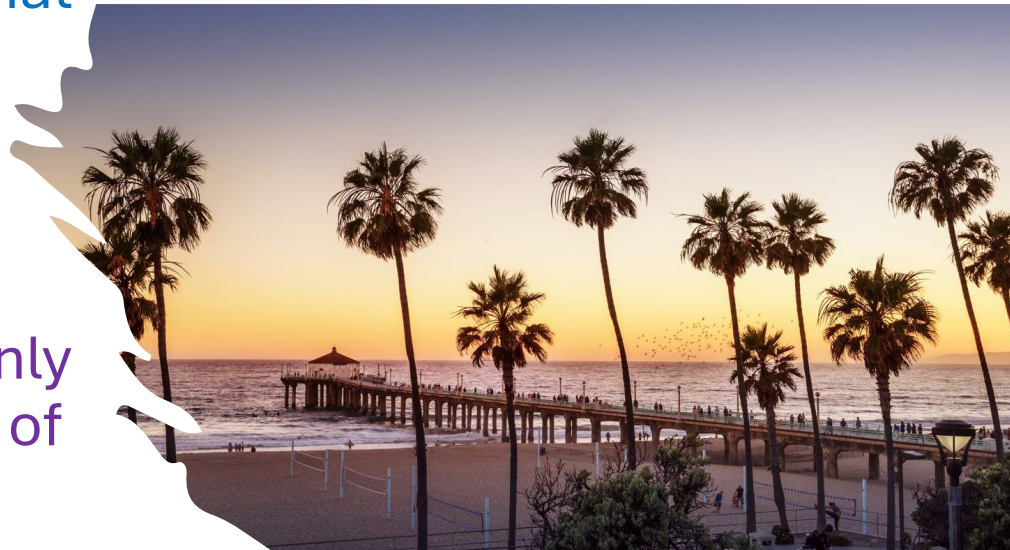
Time spent in Sentencing: 7 days

Data: the national statistics for the Criminal court statistics quarterly (April - June 2023)

But ..Can a weather forecast trained on Californian data predict a storm in the UK?

Learning only from data makes it hard to make decisions away from that data

Would you trust a computer to make a judicial judgement if only trained on some types of cases?





Also:

Some images are hard to
classify correctly



Machines can get it wrong

Identified as a 45mph speed sign



“panda”

57.7% confidence

+ .007 ×



noise

=



“gibbon”

99.3% confidence

And can get confused by an
‘adversarial attack’

Study this as a conditioning
problem in numerical analysis

Machine learning models can memorise data: our privacy is at risk

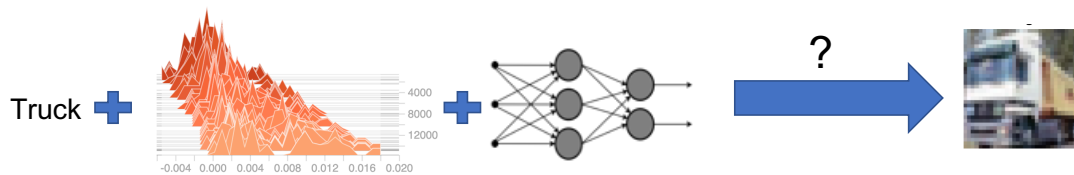
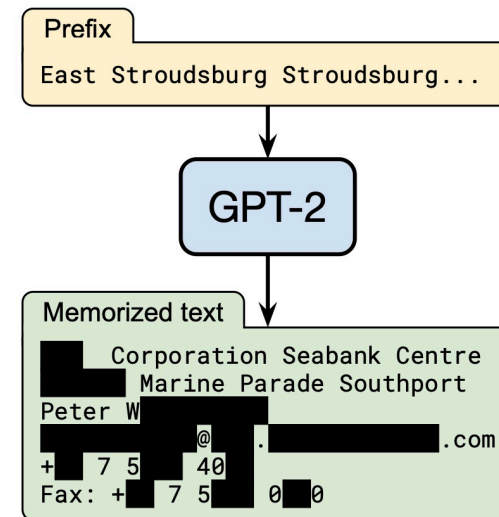
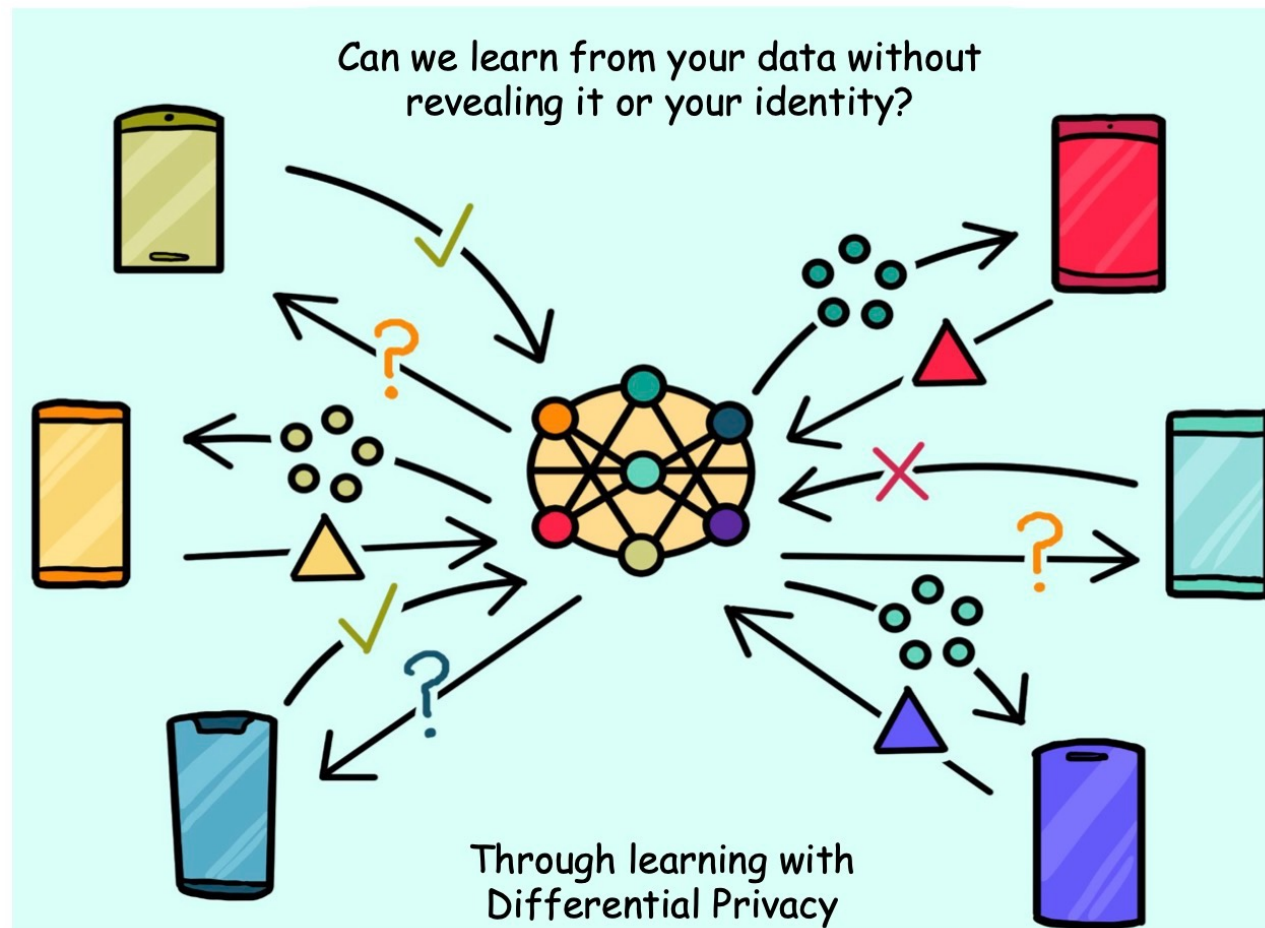


Image classification models can **memorise their training images** through the training updates (i.e. gradients).



Language models can also **memorise their training** can be extracted using text prompts.

One way to mitigate the privacy risk: training with differential privacy



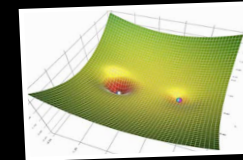
Maths is our best way of explaining how AI is working and answering questions about its limitations and its possibilities

Computational Mathematics

Work by 19th century mathematicians **Charles Babbage** and **Ada Lovelace**, and 20th century mathematicians **Alan Turing** and **John Von Neumann** leads to the invention of the modern computer



20th century developments in the **mathematics of optimisation** led to the rapid growth of machine learning



Statistics and Data Science

19th century mathematicians **Thomas Bayes** and **Carl Friedrich Gauss** transformed the way that we understand and manipulate data



$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

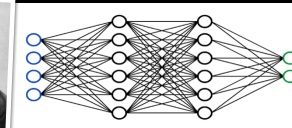
$$\theta^* = \underset{\theta \in \Theta}{\operatorname{argmin}} \left[\sum_{i=1}^N (f_{\theta}(x_i) - y_i)^2 + R(\theta) \right]$$

This mathematics is used in medical imaging



Algebra and Calculus

Matrices, invented by 19th century mathematician **Arthur Cayley**, provide the structure for the Internet and all AI architectures



$$\mathbf{x}_{k+1} = \sigma(A_k \mathbf{x}_k + \mathbf{b}_k) \in \mathbb{R}^W$$



20th century mathematician **Kiyoshi Itô** developed stochastic calculus, the basis of generative AI

